



PHYSIOLOGICAL AND ENDOCRINOLOGICAL CHANGES DURING PREGNANCY

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GENITAL CHANGE

The body of the uterus

Height and weight(hyperplasia)

The height increases from 7.5 cm to 35 cm

The weight increases from 50 g to 1000 g at term

Uterine ligaments

Show hypertrophy

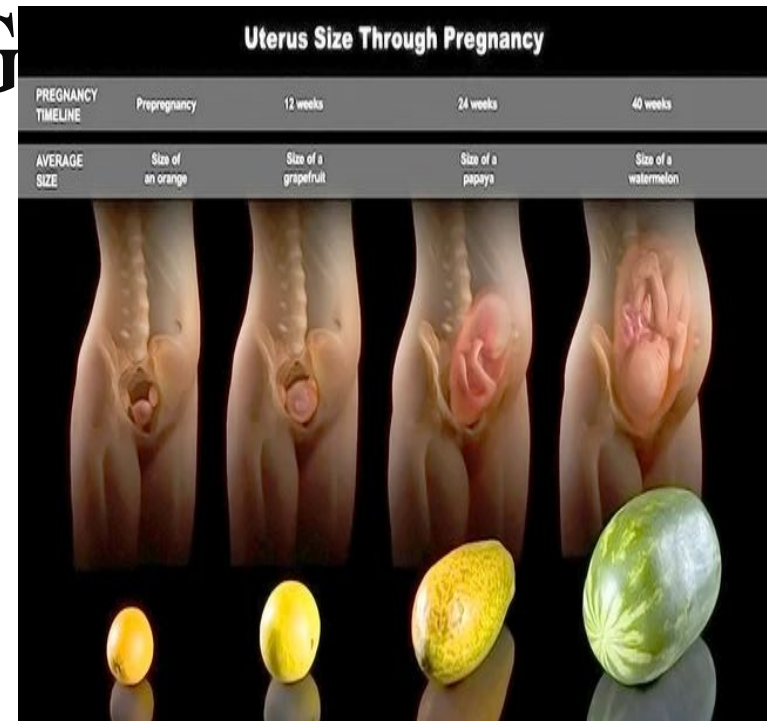
-Dextro-rotation

The uterus is tilted and twisted to the right in 80% of cases

Lower uterine segment(LUS)

The LUS is formed from the isthmus

Formed from the 4 th month to reach 10 cm at full term



GENITAL CHANGES

The cervix

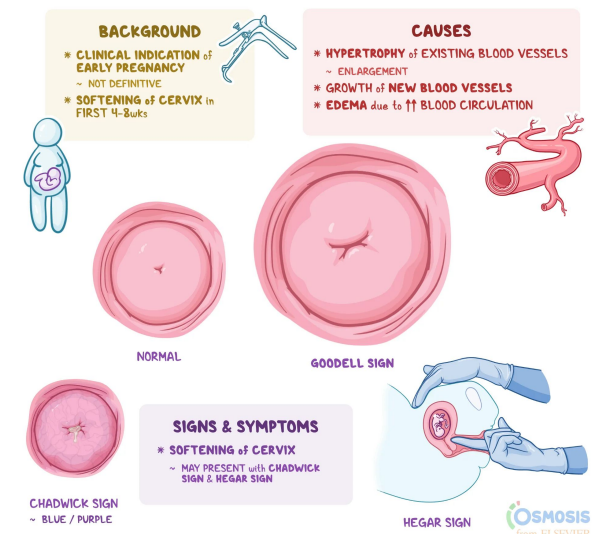
Edema and congestion, and becomes soft

Mucus plug(operculum):cervical mucus closing the cervical canal

Increases secretion from its glands

The vulva

Shows increased vascularity and varicosities





GENITAL CHANGES

The vagina

Shows increase vascularity: soft ,moist and bluish

Distension of vagina at birth

The ovary

Shows increased vascularity and size

One ovary contains the corpus luteum

Pelvic ligaments

Relaxation of the ligaments

Relaxation of the pelvic joints

The pelvis become more mobile and increases in capacity

BREAST CHANGES

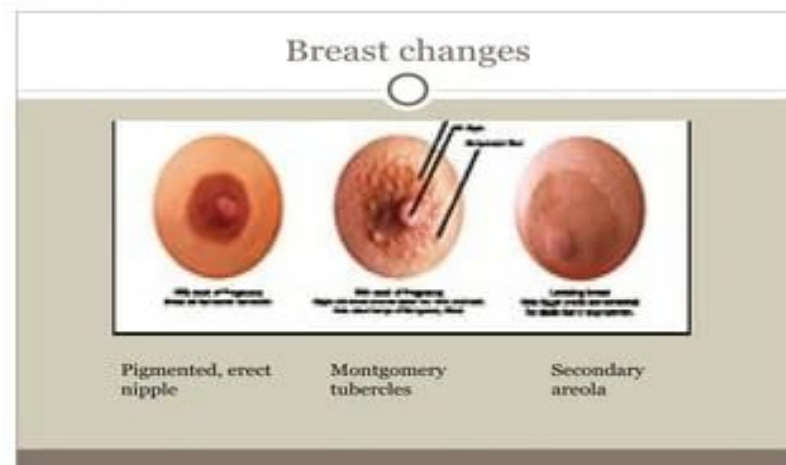
Increased size and vascularity: warm , tense and tender

Increased pigmentation of the nipple and areola

Secondary areola a(light pigmentation around the primary areola)

Montgomery tubercles appear on the areola(dilated sebaceous glands)

Colostrum like fluid is expressed at the end of the third month



SKIN CHANGES



Pigmentation

Due to increased melanocyte stimulating hormone:

Linea nigra :pigmentation of the linea alba ,more marked below the umbilicus

Chloasma gravidarum: Butterfly pigmentation of the face (mask of pregnancy)

Striae gravidarum

Stretch of the abdominal wall ,rupture of the subcutaneous elastic fibers, pink lines in flanks , become white after labor



WEIGHT INCREASE

There is an increase weight of approximately 12.5 kg at term.

The main increase occurs in the 2 nd half of the pregnancy 0.5 kg/week

Causes:

Growth of the conceptus

Enlargement of the maternal organs

Maternal storage of fat

Increase in maternal blood and
interstitial fluid

Pregnancy Weight Gain Breakdown

Baby.....	7.5 lbs
Enlarging uterus.....	2.0 lbs
Placenta.....	1.5 lbs
Amniotic fluid.....	2.0 lbs
Breast enlargement.....	2.0 lbs
Extra blood & fluid volume...	8.0 lbs
Extra fat reserves.....	7.0 lbs
Total.....	30.0 lbs

Note: All amounts are approximate

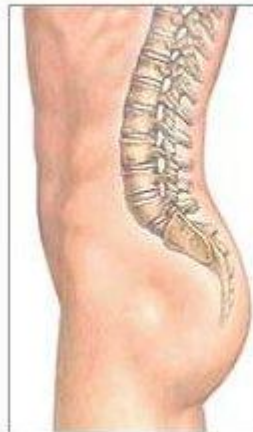


SKELETAL CHANGES

Increased lumbar lordosis

Relaxation of pelvic joints and ligaments due to progesterone and relaxin

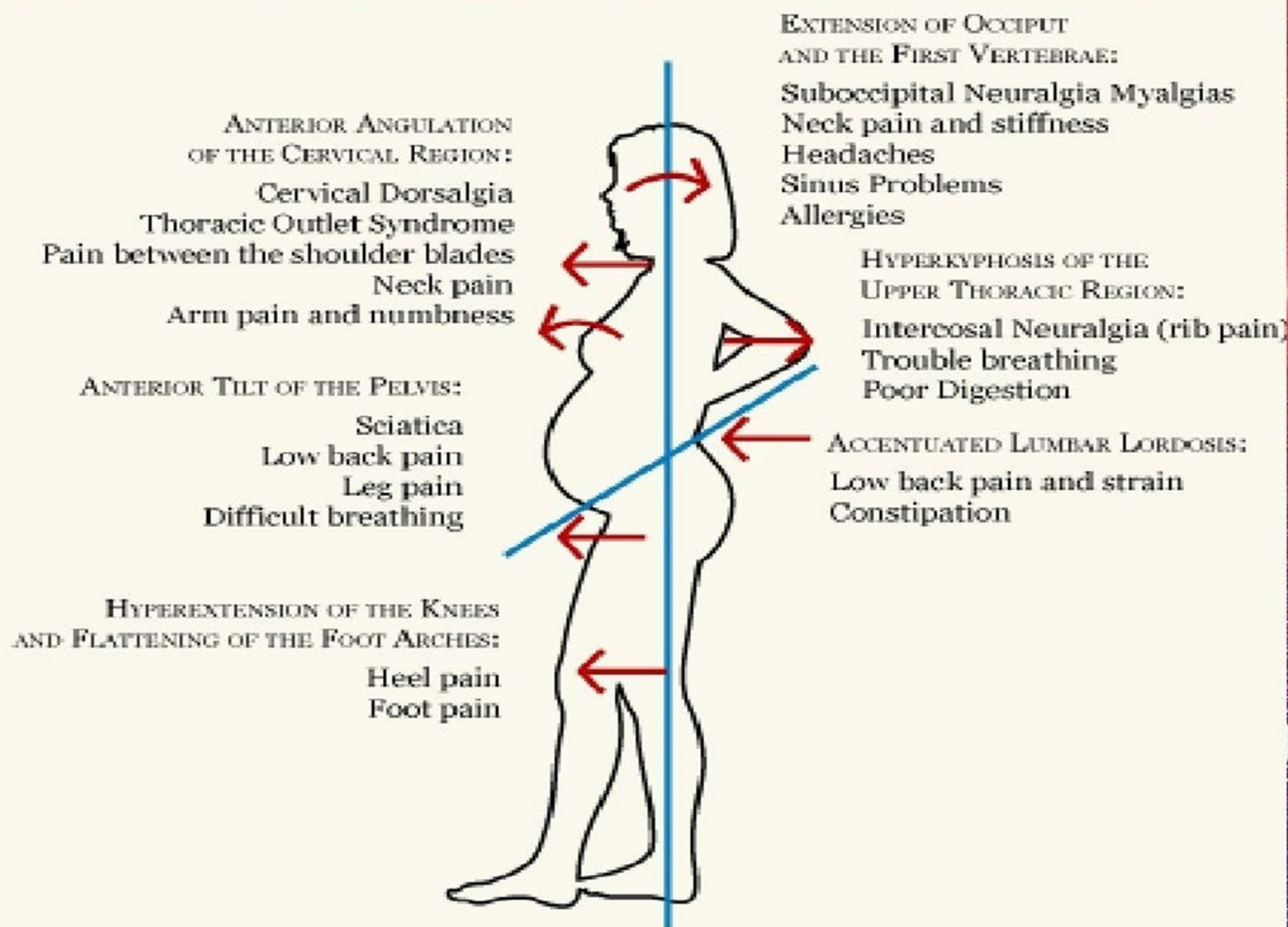
Normal spine



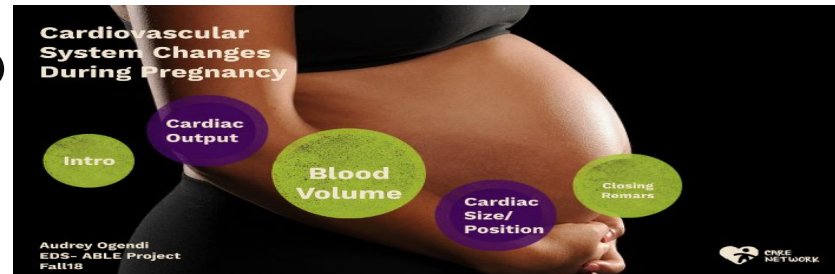
Lordosis of the spine



Exaggerated
lumbar
curve



CARDIOVAS CHANGES



Changes in the cardiovascular system in pregnancy are profound and begin early in pregnancy, such that by eight weeks' gestation, the cardiac output has already increased by 20%. The primary event is probably peripheral vasodilatation

Peripheral vasodilation leads to a 25–30% fall in systemic vascular resistance, and to compensate for this, cardiac output increases by around 40% during pregnancy.

This is achieved predominantly via an increase in stroke volume, but also to a lesser extent, an increase in heart rate

CARDIOVASCULAR CHANGES

There is a profound effect of maternal position towards term upon the haemodynamic profile of both the mother and fetus.

In the supine position, pressure of the gravid uterus on the inferior vena cava (IVC) causes a reduction in venous return to the heart and a consequent fall in stroke volume and cardiac output.

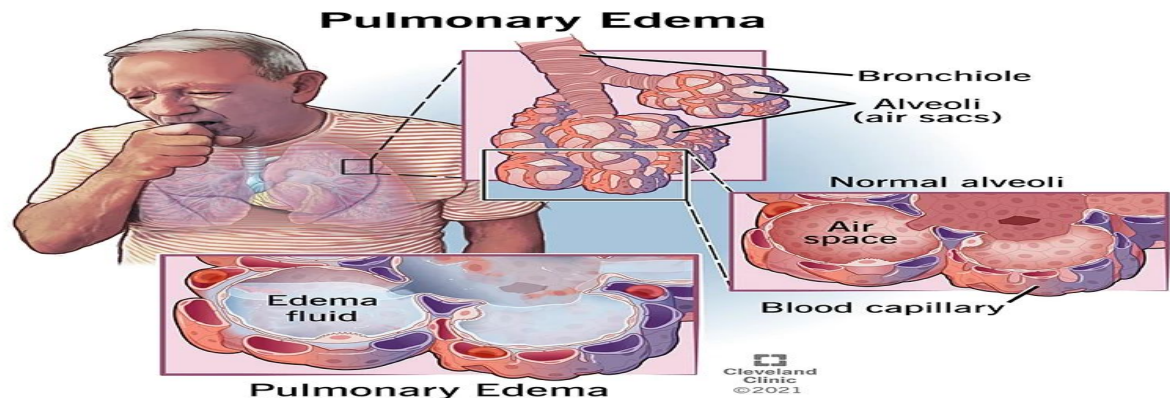
Turning from the lateral to the supine position may result in a 25% reduction in cardiac output



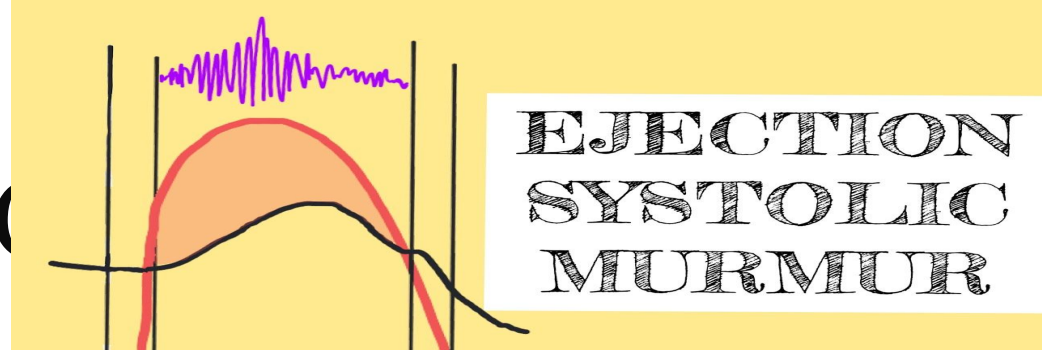
CARDIOVASCULAR CHANGES

Although both blood volume and stroke volume increase in pregnancy, Pulmonary vascular resistance (PVR), like systemic vascular resistance (SVR), decreases significantly in normal pregnancy.

Although there is no increase in pulmonary capillary wedge pressure (PCWP), serum colloid osmotic pressure is reduced by 10–15%. The colloid osmotic pressure/pulmonary capillary wedge pressure gradient is reduced by about 30%, making pregnant women particularly suscep



CARDIOVASCULAR CHANGES



The above physiological changes lead to changes so Pulmonary oedema will be precipitated if there is either an increase in cardiac pre-load (such as infusion of fluids) or increased pulmonary capillary permeability (such as in pre-eclampsia) or both

cardiovascular examination that may be misinterpreted as pathological by those unfamiliar with pregnancy. Changes may include a bounding or collapsing pulse and an ejection systolic murmur, present in over 90% of pregnant women

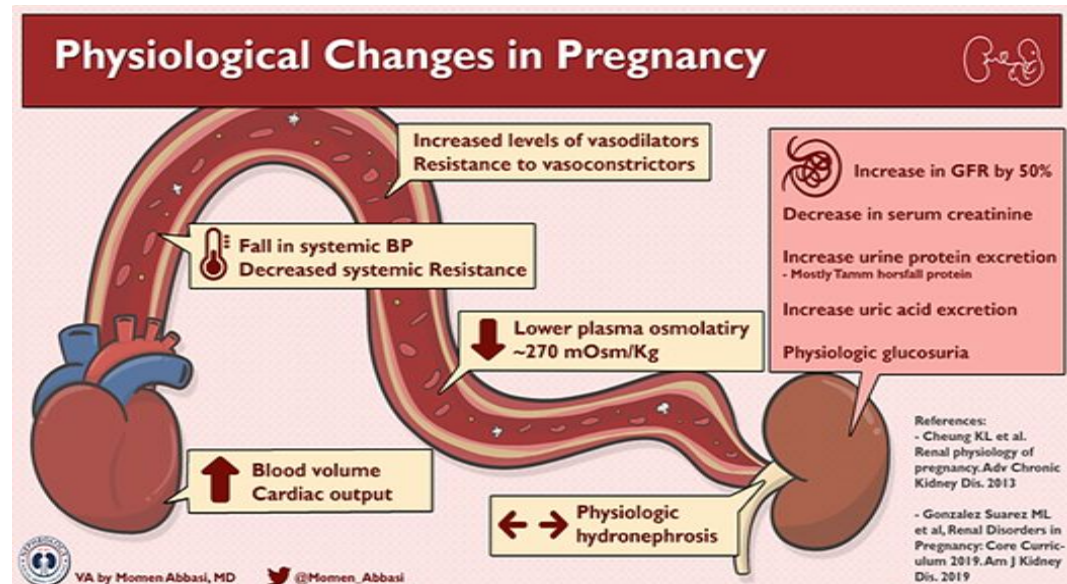


URINARY CHANGES

Changes in renal function roughly parallel those in cardiac function. Glomerular filtration rate (GFR) increases 30 to 50%, peaks between 16 and 24 weeks gestation, and remains at that level until nearly term, when it may decrease slightly

. Renal plasma flow increases in proportion to GFR. As a result, blood urea nitrogen (BUN) decreases, usually to < 10 mg/dL (< 3.6 mmol urea/L), and creatinine levels decrease proportionally to 0.5 to 0.7 mg/dL (44 to 62 micromole/L).

Marked dilation of the ureters (hydroureter) is caused by hormonal influences (predominantly progesterone) and by backup due to pressure from the enlarged uterus on the ureters, which can also cause hydronephrosis. Postpartum, the urinary collecting system may take as long as 12 weeks to return to normal





Postural changes affect renal function more during pregnancy than at other times; i.e, the supine position increases renal function more, and upright positions decrease renal function more.

Renal function also markedly increases in the lateral position, particularly when lying on the left side; this position relieves the pressure that the enlarged uterus puts on the great vessels when pregnant women are supine. This positional increase in renal function is one reason pregnant women need to urinate frequently when trying to sleep.



GASTRO-INTESTINAL CHANGES

Increased salivation (ptyalism)

Taste is often altered very early in pregnancy

Increase appetite and thirst □ frequent small snacks

Heart burn (reflux oesophagitis) relaxation of the cardiac sphincter due to progesterone and relaxin

hyperemesis gravidarum, morning sickness in 50%

Hydrochloric acid production decreases; thus ,peptic ulcer disease is uncommon during pregnancy

Constipation, reduced gut motility due to progesterone , as pregnancy progresses, pressure from the enlarging uterus on the rectum and lower portion of the colon causes constipation



GASTRO-INTESTINAL CHANGES

Liver

- hepatic synthesis of albumin, plasma globulin and fibrinogen increases
- Total hepatic synthesis of globulin increases stimulated by estrogen
- Hormone-binding globulins rise
- Gall bladder** increases in size and empties more slowly
- Relaxation of gall bladder increases the tendency of stone formation
- Cholestasis is almost physiological
- Secretion of bile is unchanged



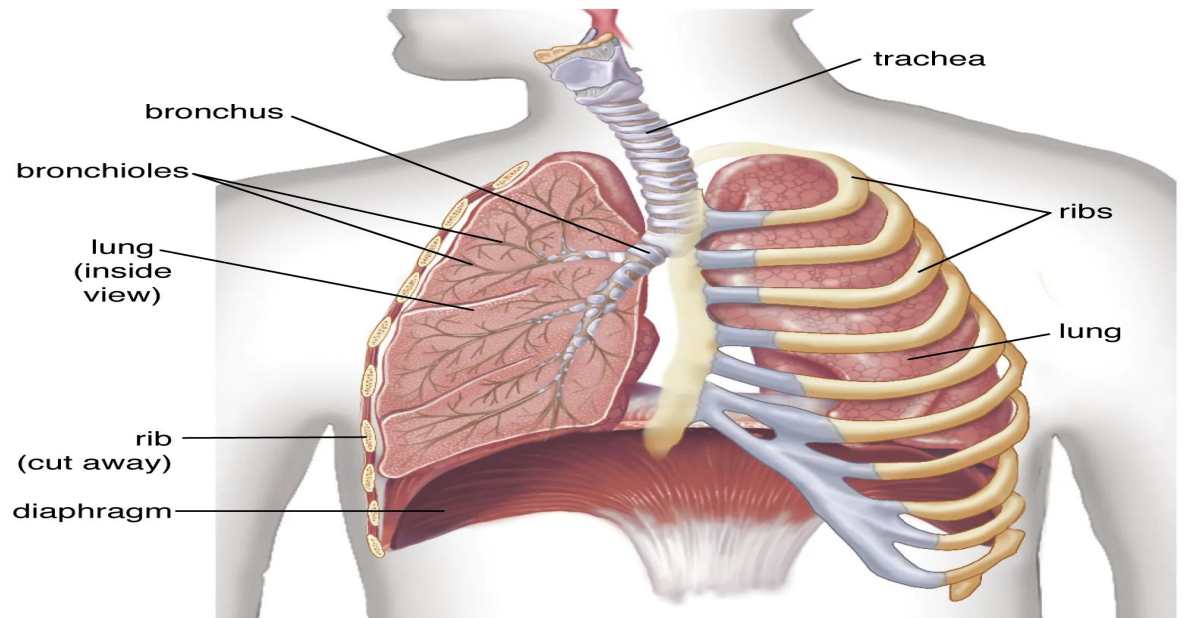
RESPIRATORY CHANGES

Lung function changes

partly because progesterone increases and partly because the enlarging uterus interferes with lung expansion.

Normal pregnancy is associated with progesterone –mediated increase in minute ventilation that results in increase in blood oxygenation(i.e. increased pO_2 and decreased in carbon dioxide(reduced pCO_2))

Reduction in P_{CO_2} results in respiratory alkalosis that is fully compensated for by increased renal excretion of bicarbonate so PH remains essentially unchanged.



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Oxygen consumption increases by about 20% to meet the increased metabolic needs of the fetus, placenta, and several maternal organs.

Functional residual capacity, expiratory reserve volume, residual volume and tidal volume decrease.

Vital capacity and total lung capacity do not change.



RESPIRATORY CHANGES

Considerable hyperemia and edema of the respiratory tract occur.

Occasionally, symptomatic nasopharyngeal obstruction and nasal stuffiness occur, eustachian tubes are transiently blocked, and tone and quality of voice change.

Mild dyspnea during exertion is common, and deep respirations are more frequent

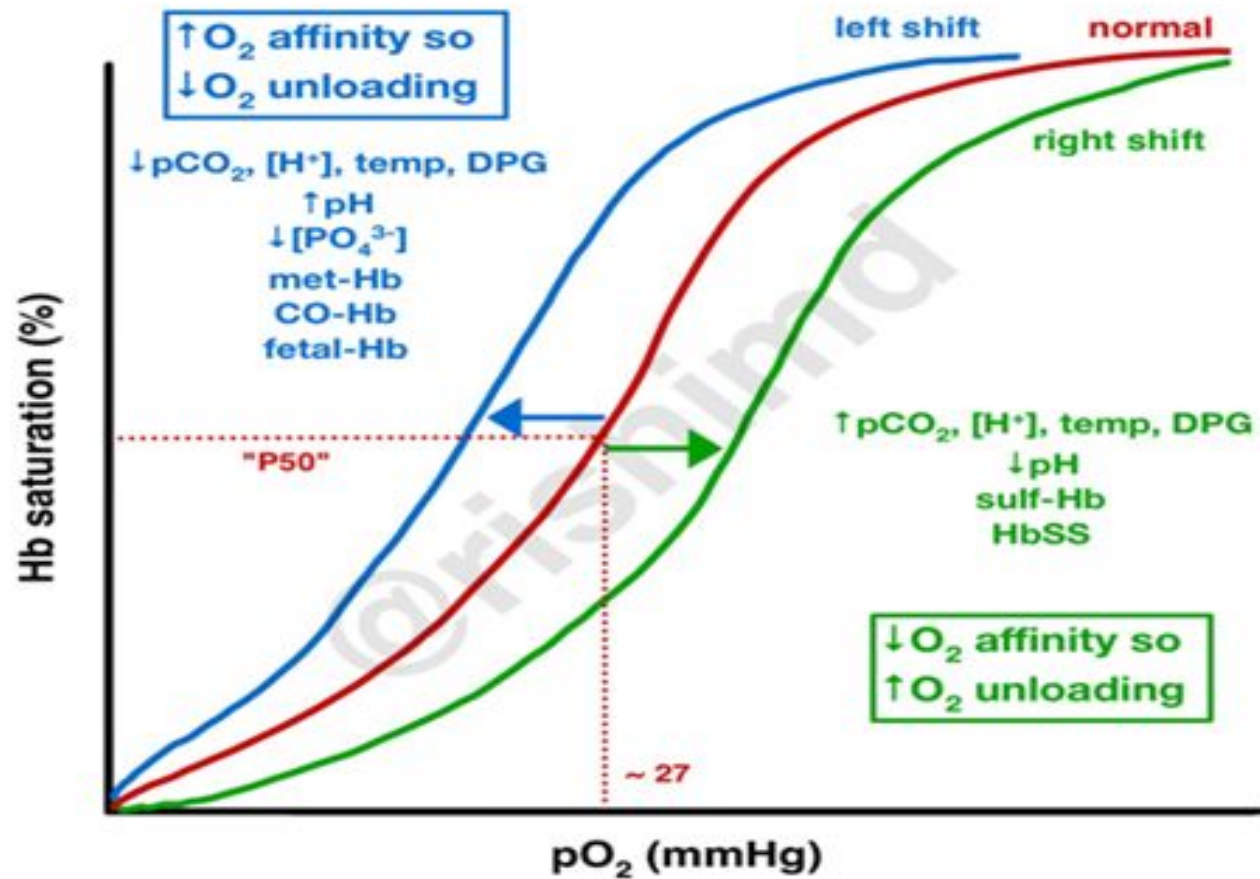


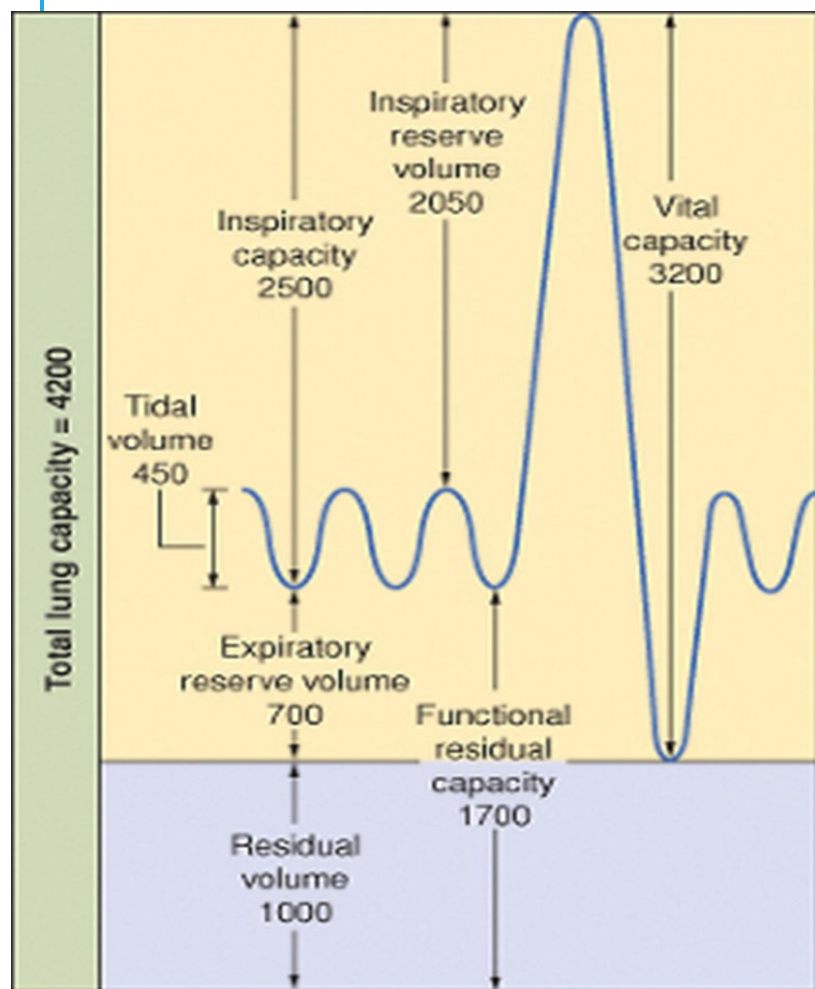
RESPIRATORY CHANGES

Pregnancy places greater demands on the cardiovascular than the respiratory system

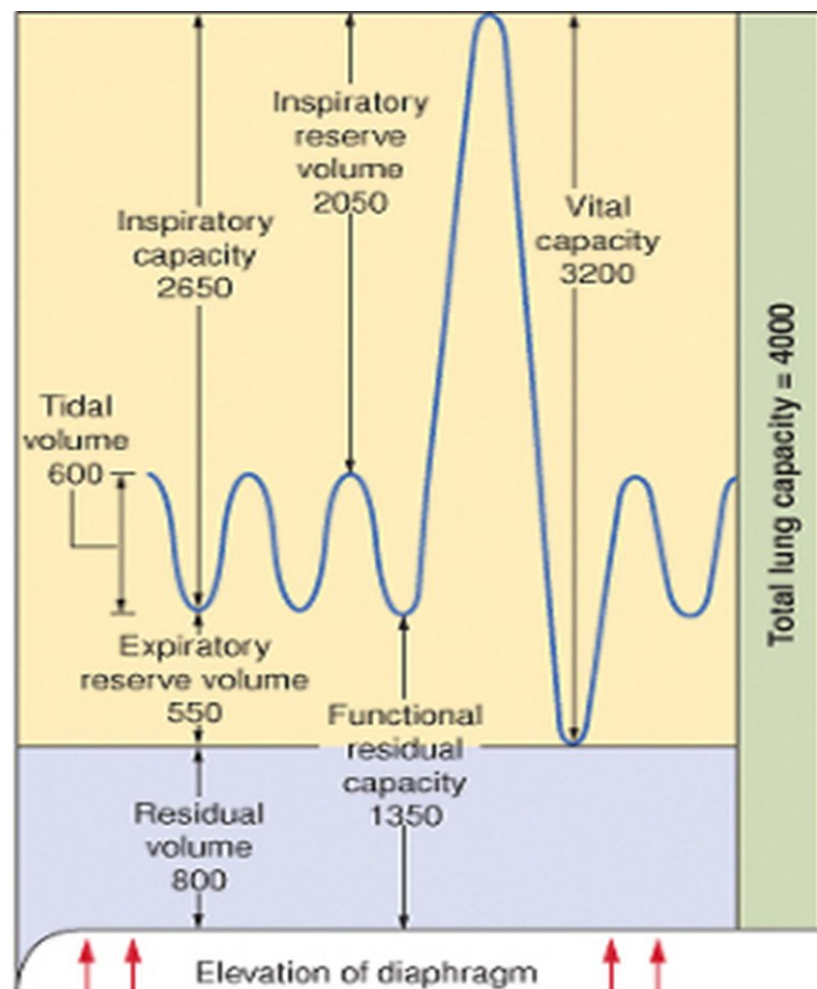
How does the dissociation curve for fetal hemoglobin compare to that for maternal hemoglobin?

Fetal hemoglobin has a higher affinity for oxygen than maternal hemoglobin at any partial pressure of oxygen.





Nonpregnant



Gravida at term



HAEMATOLOGICAL CHANGES

Total blood volume increases proportionally with cardiac output, but the increase in plasma volume is greater (close to 50%) than that in red blood cell (RBC) mass (about 25%);

thus, hemoglobin (Hb) is lowered by dilution, from about 13.3 to 12.1 g/dL.

This dilutional anemia decreases blood viscosity.

With twins, total maternal blood volume increases more (closer to 60%)



HAEMATOLOGICAL CHANGES

White blood cell count (WBC) count increases slightly to 9,000 to 12,000x10⁹/L. Marked leukocytosis ($\geq 20,000 \times 10^9/L$) occurs during labor and the first few days postpartum.

Iron requirements increase by a total of about 1 g during the entire pregnancy and are higher during the 2nd half of pregnancy—6 to 7 mg/day.

HEMATOLOGICAL CHANGES DURING PREGNANCY



- Plasma volume

Increases by 40-45%

- Red cell mass

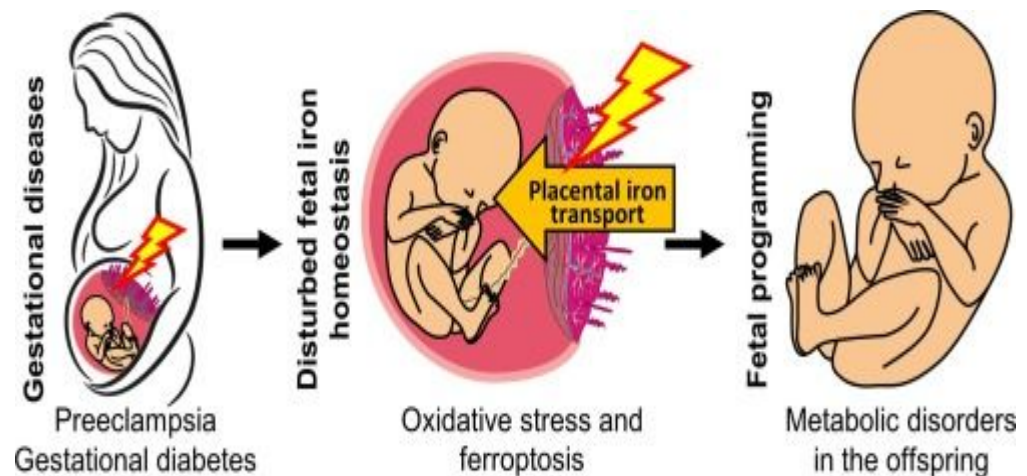
Increases by 15- 20%

Disproportionate increase b/w the plasma volume and red cell mass results in physiological anemia of pregnancy (Hb 10gm%, PCV 30 , RBC count- 3.2mill, PS)

HAEMATOLOGICAL CHANGES

The fetus and placenta use about 300 mg of iron, and the increased maternal RBC mass requires an additional 500 mg.

Excretion accounts for 200 mg. Iron supplements are needed to prevent a further decrease in Hb levels because the amount absorbed from the diet and recruited from iron stores (average total of 300 to 500 mg) is usually insufficient to meet the demands of pregnancy





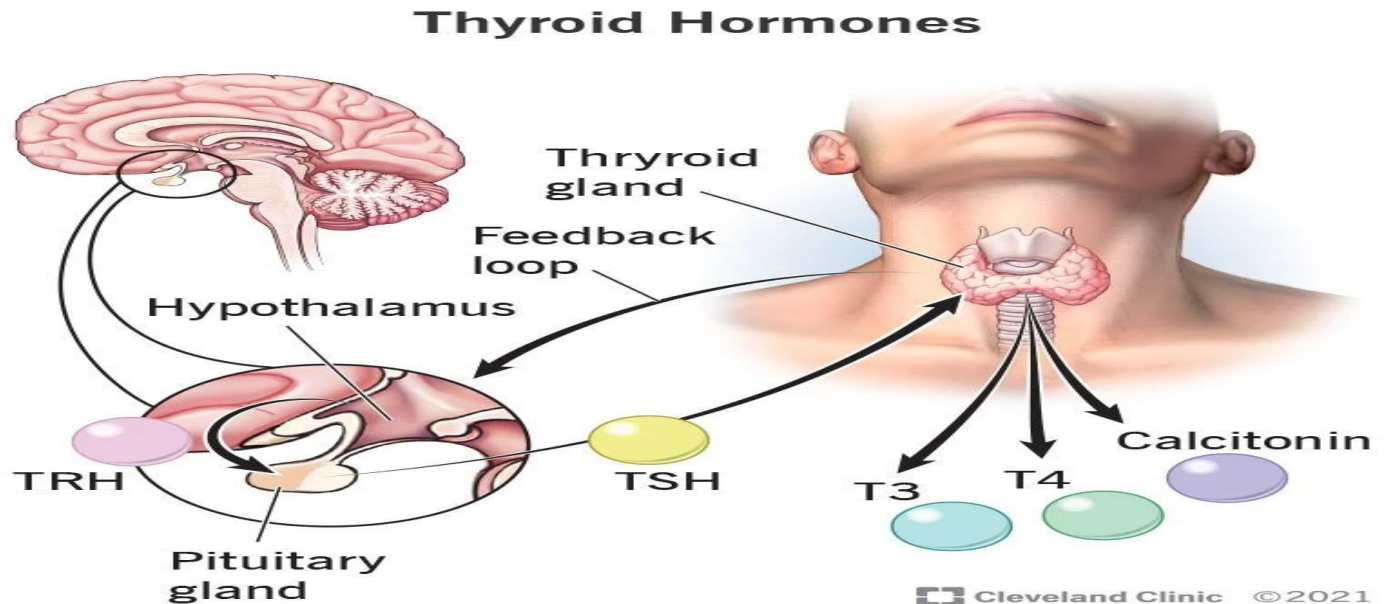
HAEMATOLOGICAL CHANGES

Coagulation

- factors VII, VIII and X rise
- absolute plasma fibrinogen doubles
- antithrombin III falls
- erythrocyte sedimentation rates increase
- protein C unchanged
- protein S concentrations, co-factor of protein C, fall in 1st and 2nd trimesters
- plasma fibrinolytic activity decreases during pregnancy & labor
 - returns to normal values within an hour of delivery of placenta

ENDOCRINAL CHANGES

Pregnancy alters the function of most endocrine glands, partly because the placenta produces hormones and partly because most hormones circulate in protein-bound forms and protein binding increases during pregnancy.





ENDOCRINAL CHANGES

Pituitary

- anterior pituitary increases in size and activity
- posterior pituitary releases oxytocin on the onset of labor

Thyroid

- increases in size and activity: physiological goiter
- most pregnant women are euthyroid
- thyroid binding globulin concentrations double (not other thyroid binding proteins)
- total T3 and T4 are increased (not the free T3 and T4)

Parathyroid

- increases in size and activity to regulate calcium metabolism



ENDOCRINAL CHANGES

Adrenals

-increase in size and activity ,total cortisol is increased (free cortisol unchanged)

Placental hormones

Progesterone

-produced by the corpus luteum ,levels rise steadily during pregnancy, output reaches 250mg/day

-actions:

Colon activity reduced, nausea, constipation

Reduced bladder and ureteric tone

Diastolic pressure reduced, venous dilation, raises temperature



ENDOCRINAL CHANGES

Placental hormones

Oestrogens

Source:

- ovary in early pregnancy

- later, oestrone and oestradiol produced by the placenta Increased a hundredfold

- oestriol produced by the placenta and fetal adrenals increased thousandfold

Levels: output of oestrogens reaches a maximum of at least 30-40mg/day

oestriol accounts 85%

levels increase up to term



ENDOCRINAL CHANGES

Placental hormones

Oestrogens

Possible actions:

- induce growth of uterus and control its function
- responsible for the development of breasts (with progesterone)
- alter chemical constitution of connective tissue, become more pliable
- cause water retention
- reduce sodium excretion

INCORRECT UPRIGHT POSTURE

HEAD

Chin pushing forward. Eyes focus down

SHOULDERS AND CHEST

Slouching constricts the ribcage, makes breathing more difficult and causes indigestion

ABS, BUTT & UTERUS

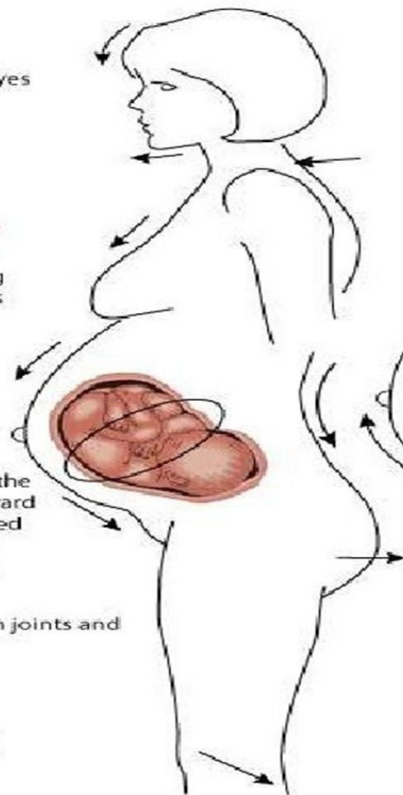
Slack muscles allow out the back and tilts pelvis forward causing backache, strained abdominals, and excess pressure on the bladder.

KNEES

If pressed back you strain joints and push pelvis forward

FEET

Weight on inner borders strains arches and calves causing leg aches.



CORRECT UPRIGHT POSTURE

HEAD

Lift through the crown of the head and keep chin lifted and ears in line with neck.

SHOULDERS AND CHEST

Draw shoulders back and down while you lift the rib cage up.

ABS, BUTT & UTERUS

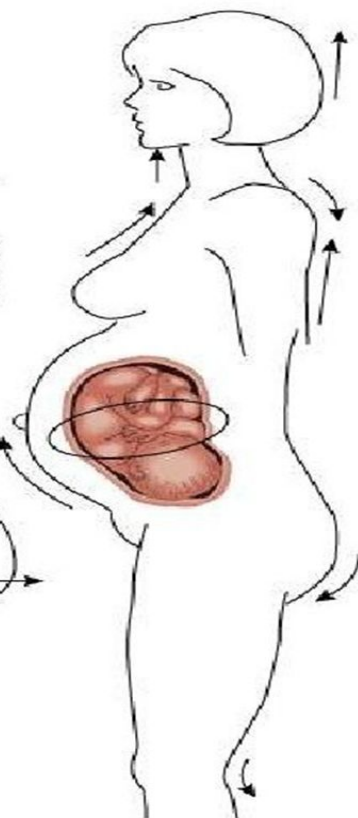
Contract abdominals to support baby, tuck butt under and tilt pubic bone slightly forward to center pelvic bowl.

KNEES

Bend knees to ease body weight over feet.

FEET

Distribute body weight over center of each foot.





METABOLIC CHANGES

Carbohydrate metabolism

- pregnancy is hyperlipidaemic and glucosuric
- after mid-pregnancy, resistance of insulin develops
- plasma glucose concentrations rise, maintained between 4.5-5.5mmol/L
- glucose crosses the placenta, the fetus uses glucose as primary energy substrate, transport occurs by carrier mediated mechanism
- the insulin resistance is endocrine-driven, via increase in cortisol hPL
- concentrations of glucagons and the catecholamines are unaltered



METABOLIC CHANGES

Carbohydrate metabolism

- carbohydrate deposited in the liver as glycogen
- some escapes to general circulation
- portion metabolized by the tissues:

Converted to depot fat

Stored as muscle glycogen

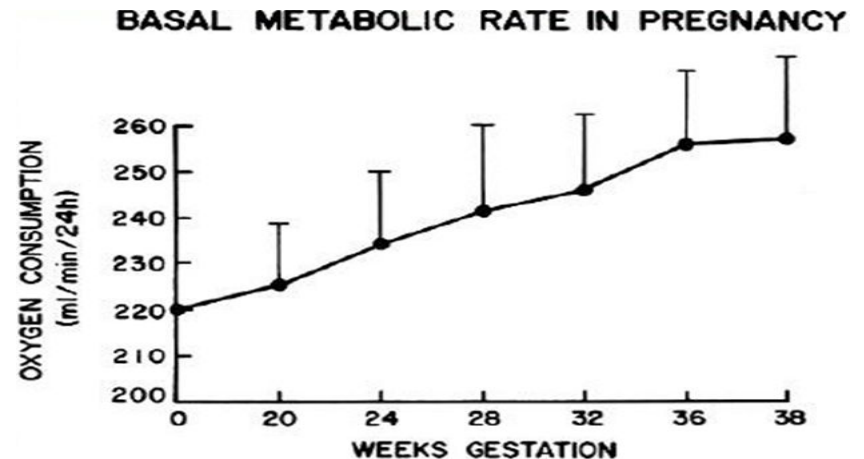
- first noticeable change occurs in blood sugar
- tested by giving a load of oral glucose (glucose tolerance test)
- the blood sugar, after meal, remains high facilitating placental transfer

METABOLIC CHANGES

Carbohydrate metabolism

- with increased placental production of steroid, less glycogen deposited in liver and muscles
- the effect of fasting is pronounced in pregnancy overnight fast of 12hrs

hypoglycemia, production of ketone bodies





METABOLIC CHANGES

Protein metabolism

- positive nitrogen balance
- on average 500g of protein retained by the end of pregnancy
- blood and urine urea are reduced

Fat metabolism

- by 30 weeks, 4kg are stored in form of depot fat in the abdominal wall, back and thighs
- modest amount in breasts

Thank
you!